What are Packages?

Packages are used to group or package multiple modules that work together to solve a problem

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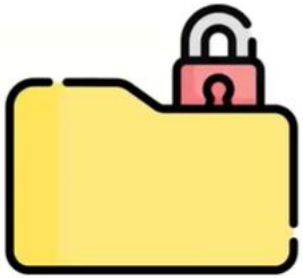
What are Packages?

For example if you wish to write code that can compress a file, you may end up writing several modules dealing with different aspects of the compression process



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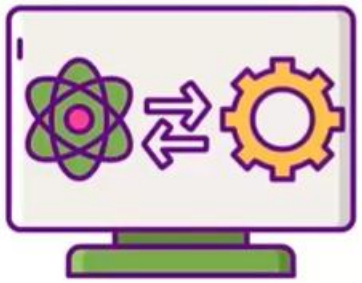
What are Packages?

However if you wish to group these modules under a single name and apply some structure to it to make it reusable then you can use packages in python

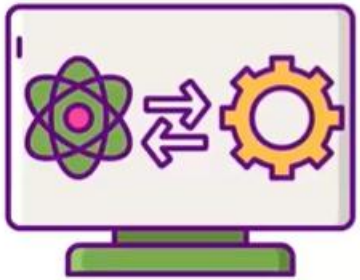
How to create a package

Creating a package is as simple as putting a file named __init__ in the directory containing related modules



How to create a package

This __init__ file can be empty or contain code that sets the value of __all__ global variable

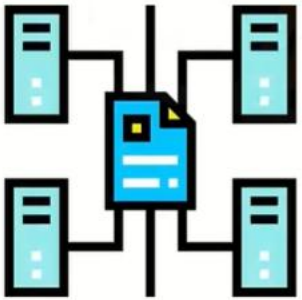


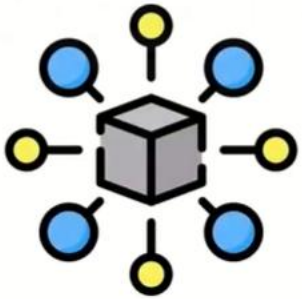
How to create a package

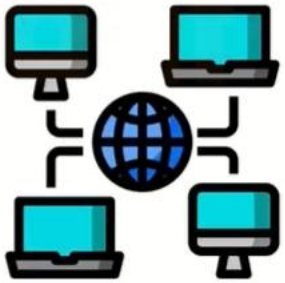
This is done because a module may contain several submodules, and the developer may not wish for any user to be able to import all of those as it may cause name conflict and other side effects

Using external packages

Many times we may wish to use packages created by other developers







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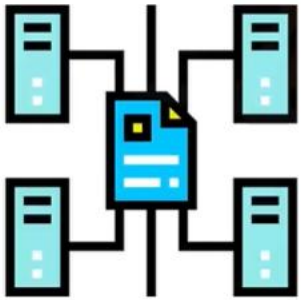
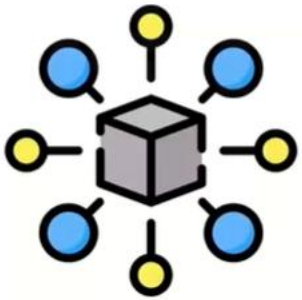
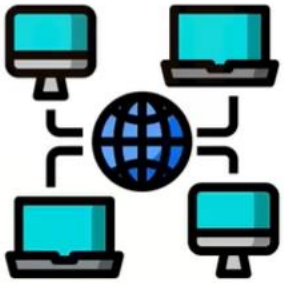
Using external packages

These packages may be performing tasks which we do not know how to implement, for example, parsing an MP3 file







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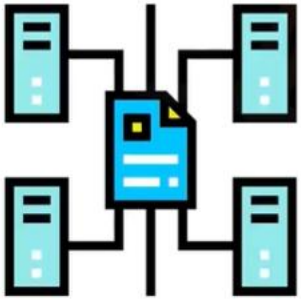
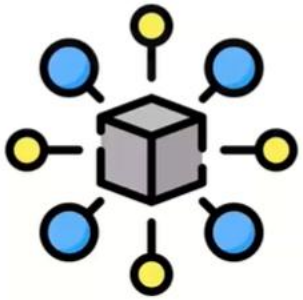
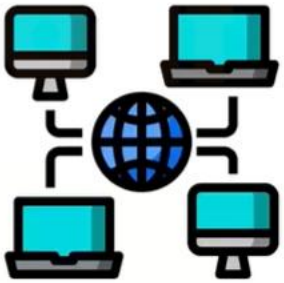
Using external packages

To use these packages in python we can use the python package manager pip

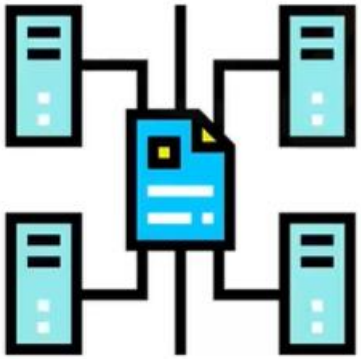


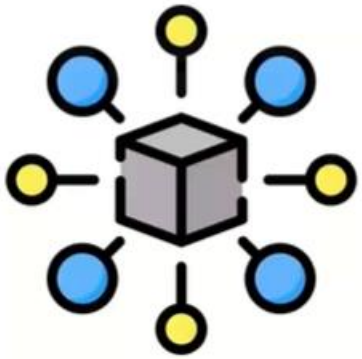




Using external packages

To install a package we can use the command pip install package_name, where package_name is the name of the package which we wish to install on our system and use





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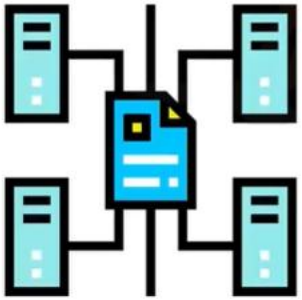
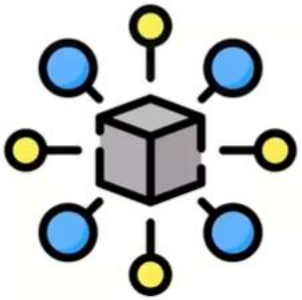
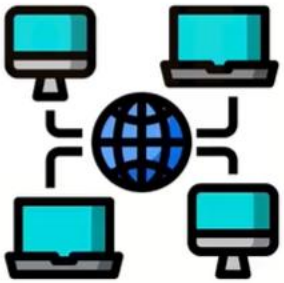
Using external packages

However, it recommended that you install these packages using pipenv







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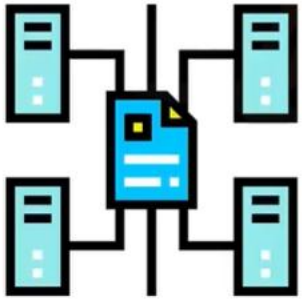
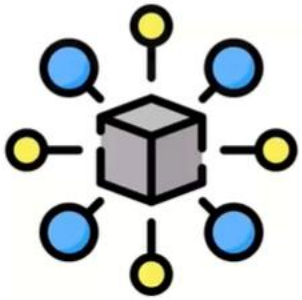
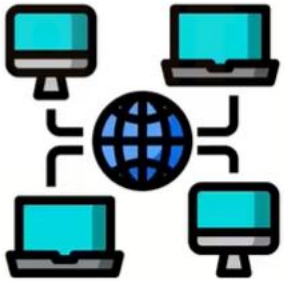
Using external packages

pipenv creates a virtual environment in which you can install your desired packages without affecting packages for installed other projects







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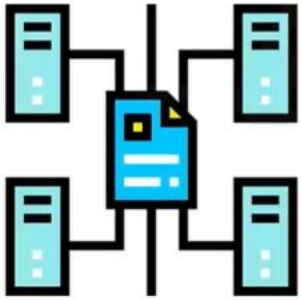
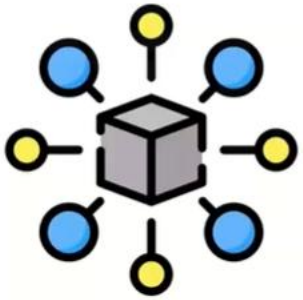
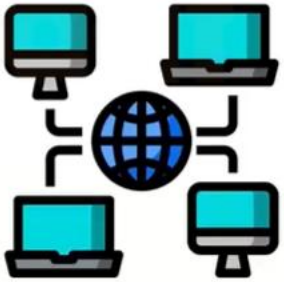
Using external packages

This is done so that if you have two projects using different versions of the same package then they can have a local copy of that package without affecting the other projects on the system







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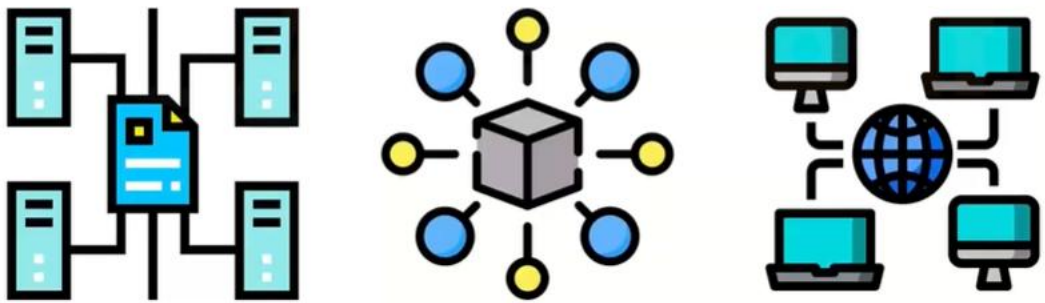
Using external packages

To use pipenv firstly create a pipenv virtual environment using the command pipenv shell in the root folder containing your python project



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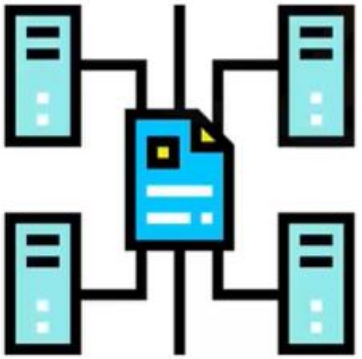
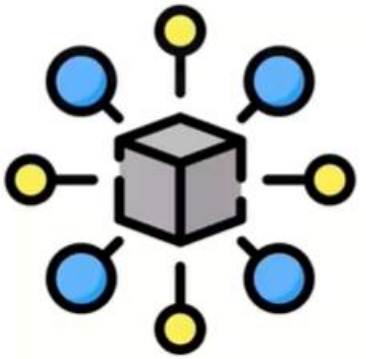
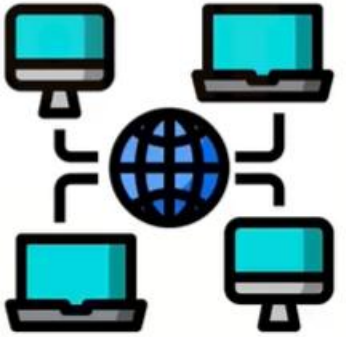
Using external packages

Then use pipenv install package_name to install and use the package







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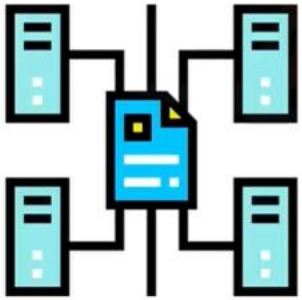
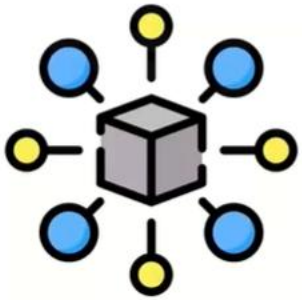
Using external packages

pipenv also creates a pipfile in the root directory so that other developers working on the project can install the same packages on their systems using the command pipenv install





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